INCH-POUND
MIL-DTL-27072/76C
w/AMENDMENT 1
20 May 2016
SUPERSEDING
MIL-DTL-27072/76C
12 March 2012

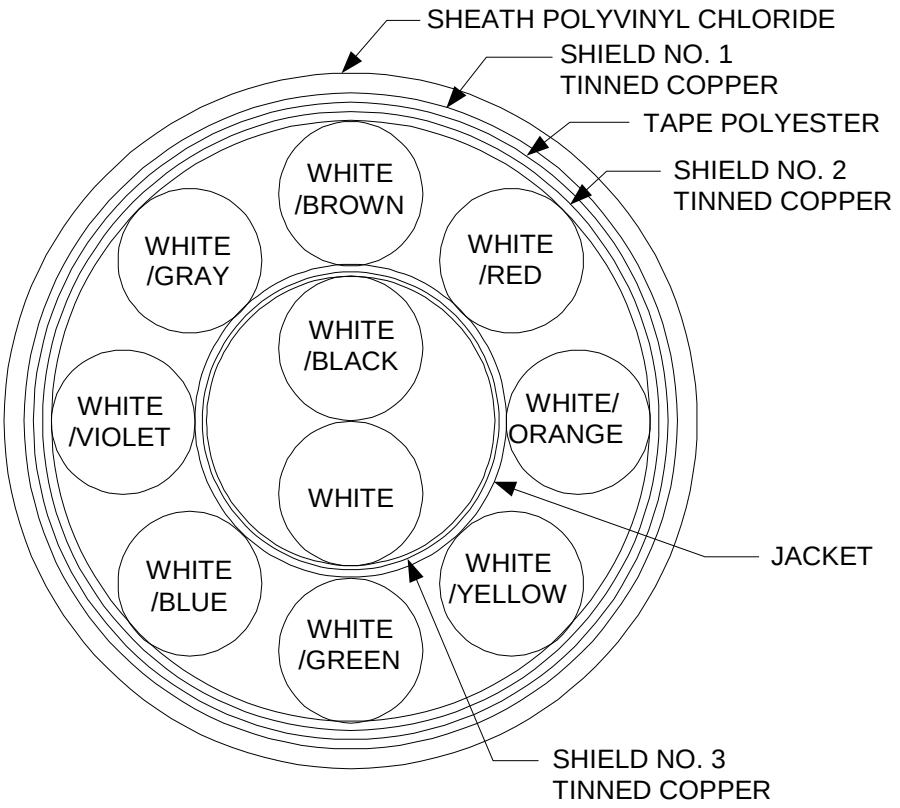
DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR (1 SHIELDED PAIR AND 8 WIRES UNDER A DOUBLE SHIELD), M27072/76SACA*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27072.

REQUIREMENTS:



NOTE: The complete Part or Identifying Number (PIN) requires a color code designator (see table I).

FIGURE 1. Cable configuration.

AMSC N/A

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:54Z
Check the source to verify that this is the current version before use.

Design and construction:	See figure 1
Number of conductors:	10
Temperature rating:	105°C
Components:	
Basic wire:	NEMA HP 7-BN-BDB* (see table I)
Conductor:	Tinned copper
Primary insulation:	Polyvinyl chloride
Colors:	White White with black tracer White with brown tracer White with red tracer White with orange tracer White with yellow tracer White with green tracer White with blue tracer White with violet tracer White with gray tracer
Jacket over insulation:	Polyamide
Color:	Natural
Filler:	Not applicable
Cabling:	
Core:	
Number of conductors:	2 (1 twisted pair)
Pitch or lay:	1.50 inches nominal
Shield Number 3:	
Material:	Tinned copper
Gauge:	36 AWG
Braid angle:	20 to 40 degrees
First layer:	
Number of conductors:	8
Pitch or lay:	3.0 inches nominal
Shield Number 2:	
Material:	Tinned copper
Gauge:	36 AWG
Braid angle:	20 to 40 degrees
Binder tape material:	Polyester (over shield number 2)
Shield Number 1:	
Material:	Tinned copper
Gauge:	36 AWG
Braid angle:	20 to 40 degrees
Jacket undershield:	Not applicable
Sheath:	
Material:	Polyvinyl chloride

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:54Z
Check the source to verify that this is the current version before use.

Wall thickness:	.040 inch (1.0 mm) nominal
Tensile strength:	In accordance with MIL-DTL-27072
Ultimate elongation:	In accordance with MIL-DTL-27072
Accelerated aging:	In accordance with MIL-DTL-27072
Color:	See MIL-DTL-27072
Finished cable:	See table I
Cold bend:	Mandrel size in accordance with MIL-DTL-27072
Dielectric withstanding voltage:	2,000 V rms
Conductor resistance:	In accordance with basic wire specification

TABLE I. Construction data.

PIN 27072/76	Basic wire except color NEMA HP 7-BN-BDB*1/,2/ (reference)				Finished cable diameter inch(mm) ±10%
	Wire size (AWG)	Number of strands	Strand size	Nominal diameter inch(mm)	
SACA	26	7	34	.044(1.12)	.297(7.54)

1/ Asterisks (*) shall be replaced by the appropriate color designator.
2/ Dimensions are in inches. Metric equivalents are given for information only.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced document. This document references MIL-DTL-27072.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 6145-2016-026)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:54Z
Check the source to verify that this is the current version before use.